

ASIF SHAHRIAR

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RESEARCH INTERESTS

- Computer Security, AI Security, Vulnerability Detection, Adversarial ML
- Behavioural Security in LLMs and LLM-Agents

EDUCATION

Bangladesh University of Engineering & Technology (BUET) 2019 - 2024

B.Sc in Computer Science & Engineering

- CGPA: **3.85/4.0** [Scholaro US-equiv: **3.91/4.0**, CSE major-only CGPA: **3.91/4.0**]
- **Thesis:** 5G Vulnerability Detection using LLMs through Zero-Shot and In-Context Learning (published in **IEEE TIFS**). Supervisor: [Prof. Md. Shohrab Hossain](#)

PEER-REVIEWED PUBLICATIONS

[1] Words Speak Louder Than Code: Investigating Cognitive Heuristics in LLM-Based Code Vulnerability Detection

Asif Shahriar, Hongyu Cai, Hadjer Benkraouda, Gang Wang, Z. Berkay Celik

IEEE Symposium on Security and Privacy (Oakland), 2027 [\[Paper\]](#) [\[GitHub\]](#)

[2] 5GPT: 5G Vulnerability Detection by Combining Zero-Shot Capabilities of GPT-4 With Domain-Aware Strategies Through Prompt Engineering

A. Shahriar, S. J. Hisham, K. M. A. Rahman, R. Islam, M. S. Hossain, R. H. Hwang, and Y. D. Lin

IEEE Transactions on Information Forensics and Security (IEEE TIFS), 2025 [\[Paper\]](#)

[3] Inceptive Transformers: Enhancing Contextual Representations through Multi-Scale Feature Learning Across Domains and Languages

Asif Shahriar, Rifat Shahriyar, M Saifur Rahman

Proceedings of EMNLP 2025 (main) [\[Paper\]](#) [\[GitHub\]](#)

[4] XLNet-CNN: Combining Global Context Understanding of XLNet with Local Context Capture through Convolution for Improved Multi-Label Text Classification

Asif Shahriar, Debojit Pandit, M Saifur Rahman

11th International Conference on Networking, Systems, and Security, 2024 (Best Paper Award) [\[Paper\]](#) [\[GitHub\]](#)

PRE-PRINTS UNDER REVIEW

[1] AgentTell: Behavioural Side-Channel Leakage in Browser-Use Agents

Asif Shahriar, Md Nafiu Rahman, Sadif Ahmed, Farig Sadeque, Md Rizwan Parvez

Under review in **ICLR 2027** [\[GitHub\]](#)

[2] A Survey on Agentic Security: Applications, Threats and Defenses

Asif Shahriar, Md Nafiu Rahman, Sadif Ahmed, Farig Sadeque, Md Rizwan Parvez

Under review in **TMLR, 2026** [\[Paper\]](#) [\[GitHub\]](#)

RESEARCH EXPERIENCE

Research Intern (Remote), PurSec Lab, Purdue University

Dec 2025 – Present

Advisors: [Prof. Z. Berkay Celik](#) (Purdue), [Prof. Gang Wang](#) (UIUC)

- Led a study showing that LLM-based vulnerability detection is susceptible to three classic cognitive biases: halo effect, framing effect and anchoring effect
- Showed that these biases can arise naturally in deployed CI/CD systems through benign non-code context, and attackers can exploit these biases to suppress detection of vulnerable code in AI-assisted review tools by up to 97%

- Resulted in a first-author paper at **IEEE S&P (Oakland) 2027**

Research Collaboration, QCRI | Remote

Aug 2025 – Present

Advisor: [Dr. Md Rizwan Parvez](#) (Scientist, QCRI)

- Led a study formally defining and investigating behavioural side-channel leakage in browser-use agents, where an agent's actions on a website reveal private information from its context, without any adversarial injection or instruction
- Built AgentTell, a benchmark of 100 tasks across 20 scenarios, and found leakage in 56–70% of tasks across six backbone models [Under review, **ICLR 2027**]
- Also led a survey on agentic security covering applications, threats, and defenses [Under review, **TMLR**]

TEACHING EXPERIENCE

Department of CSE, BRAC University

2024 – Current

Full-time Lecturer

[faculty-webpage](#)

- Artificial Intelligence (Theory and Sessional)
- Data Communications (Theory)
- Data Structures and Algorithms (Sessional)

ACADEMIC SERVICE

- **Reviewer**, EMNLP 2026
- **Reviewer**, IEEE Transactions on Cybernetics
- **Reviewer**, Cybersecurity (Springer)

TECHNICAL SKILLS

Languages: Python • Java • Javascript • C • C++ • $\text{\LaTeX}$ • Assembly

LLMs & Agents: RAG • fine-tuning • in-context learning • prompt engineering • tool-using agents • browser-use agents • coding agents • agentic security evaluation

Deep-Learning: PyTorch • TensorFlow • Neural Networks (ResNet, InceptionNet, ViT, RNN, LSTM) • Transformers • Language models

Systems & Tooling: Linux • Docker • Git • OracleDB • PostgreSQL

Networking & Security: 5G • LTE • Cryptography • EDR • NS2 • Open5GS • UERANSIM • 5GReplay • Scapy • CPPCheck • Wireshark

HONOURS, AWARDS & GRANTS

- **Best paper award** in 11th International Conference on Networking, Systems, and Security (NSysS 2024)
- **Dean's List award** for academic excellence in undergrad
- Full funding from ACL to present paper in the main conference of **EMNLP 2025**
- **Talent-pool scholarship** in HSC from Ministry of Education, Bangladesh (Merit Position: **10th**)
- **Talent-pool scholarship** in SSC from Ministry of Education, Bangladesh (Merit Position: **26th**)

REFERENCES

- **Dr. Md. Shohrab Hossain**, *undergrad thesis supervisor, co-author*
Professor, CSE, BUET | email: mshohrabhossain@cse.buet.ac.bd
- **Dr. Z. Berkay Celik**, *research supervisor, co-author*
Associate Professor, Department of Computer Science, Purdue University
email: zcelik@purdue.edu
- **Dr. Gang Wang**, *research supervisor, co-author*
Associate Professor, Computer Science, University of Illinois Urbana-Champaign (UIUC)
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